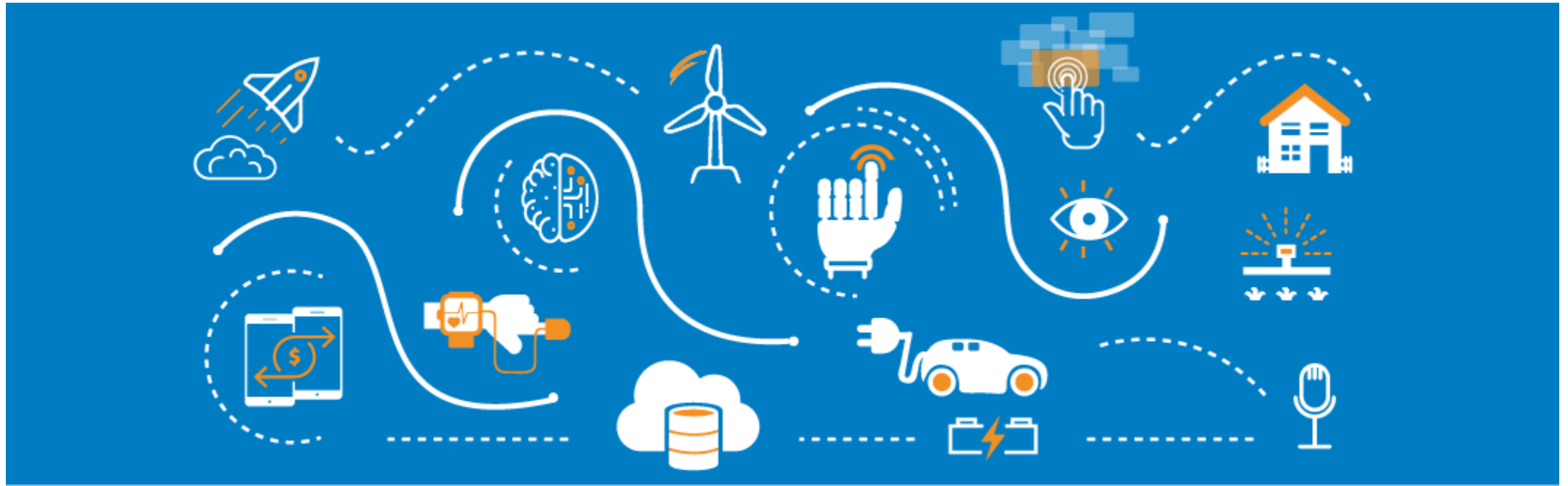




Computer Vision Part 3

Kien Tran

Contents

- 1** What is Video Analytics
- 2** Object Detection in Video Analytics
- 3** Video Analytics Applications
- 4** Demo A Video Analytic Pipeline
- 5** Project Lab

1. What is Video Analytics

1.1. Introduction

Section goals:

- What is Video Analytics
- Main Stages Video Analytics
- Video Analytics with Deep Learning

1. What Is Video Analytics?

1.2. What is Video Analytics

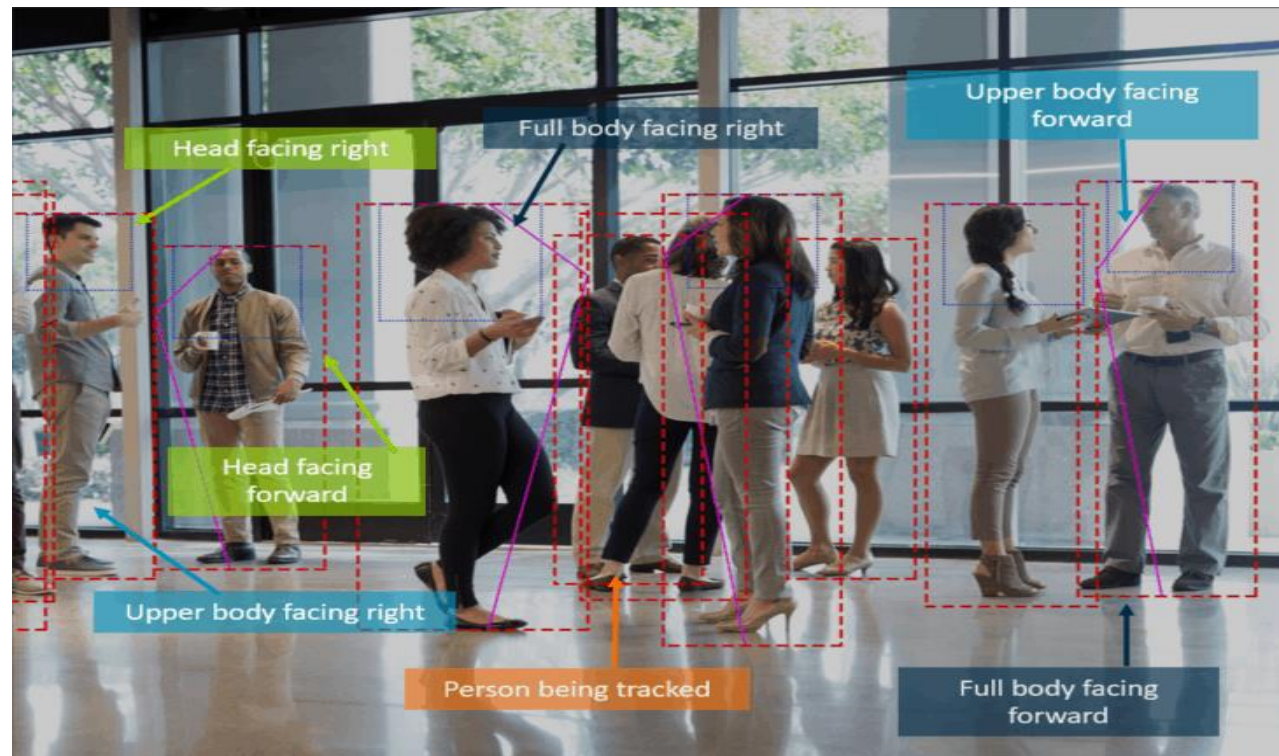
Video analytics is the process of extracting meaningful insights and information from videos using advanced computer vision and machine learning techniques



1. What Is Video Analytics?

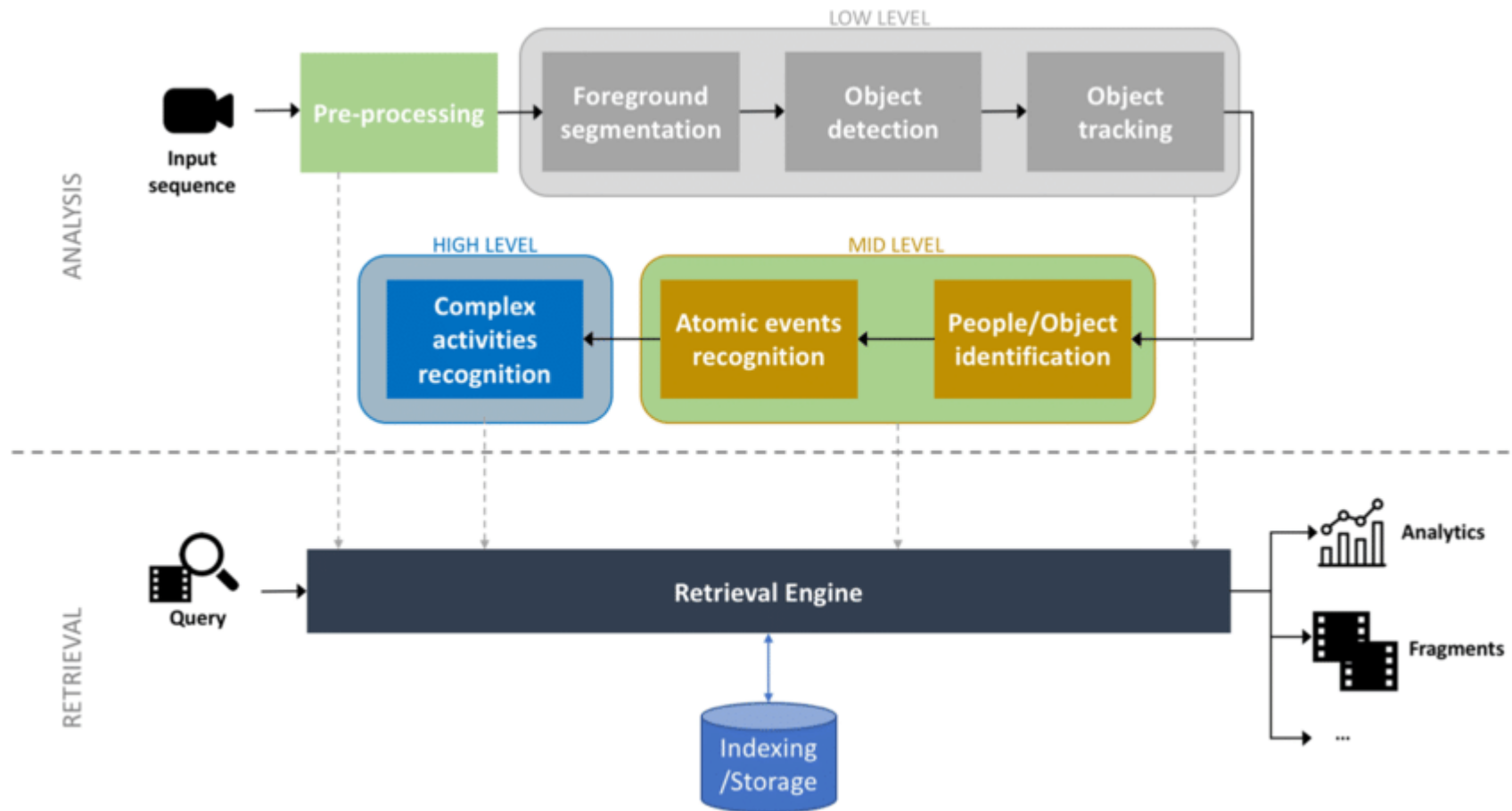
1.2. What is Video Analytics

With the growing usage of surveillance cameras, smartphones and other intelligent devices that can capture videos, there is a massive amount of untapped video data available. Video analytics helps organizations analyze this video data to extract valuable insights



1. What Is Video Analytics?

1.3. Main Stage Video Analytics



1. What Is Video Analytics?

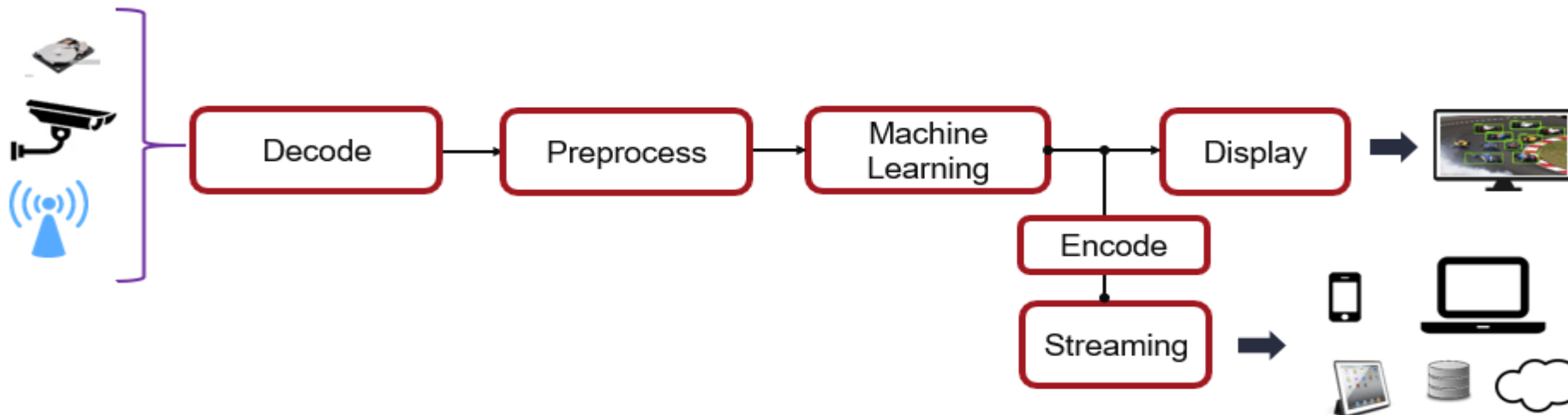
1.4. Video Analytic In Deep Learning

Traditional video analysis relied on human-defined rules which could only detect pre-programmed behaviors and had limitations.



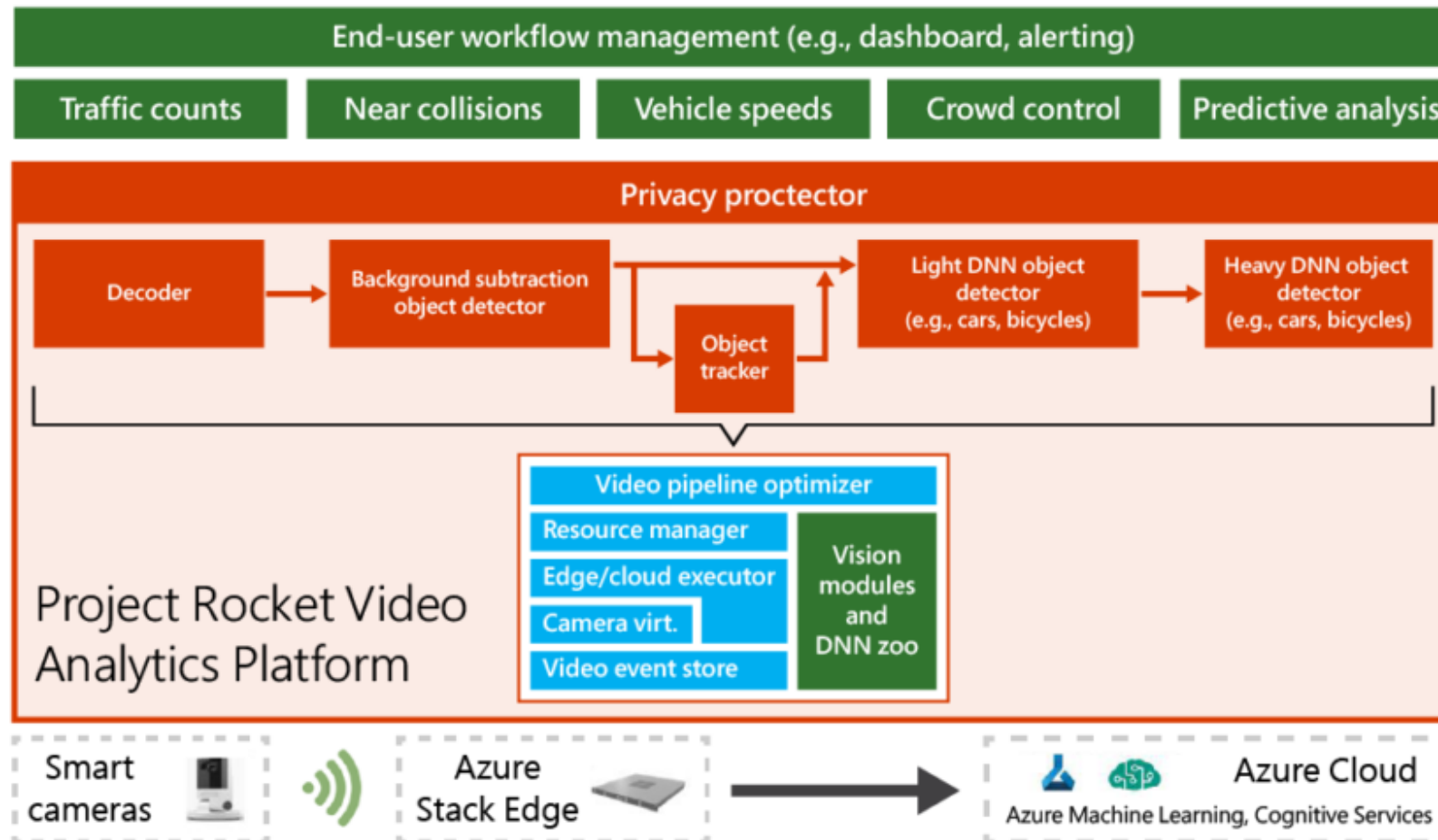
DL: Revolutionizing Video Data

- Capable of autonomous video understanding
- Deep learning algorithms are trained on massive labeled video datasets to learn the complex visual patterns and context in an end-to-end manner.
- As the systems analyze more real-world video data over time, their accuracy and capabilities improve continually



1. What Is Video Analytics?

1.4. Video Analytic In Deep Learning

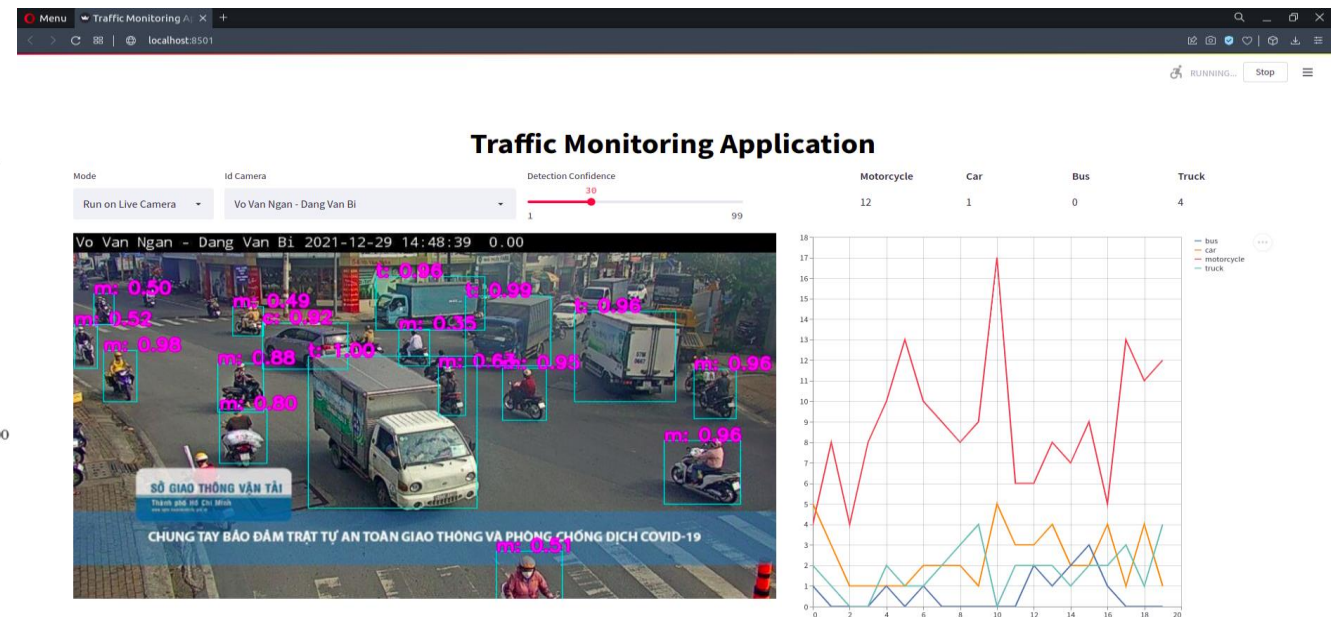
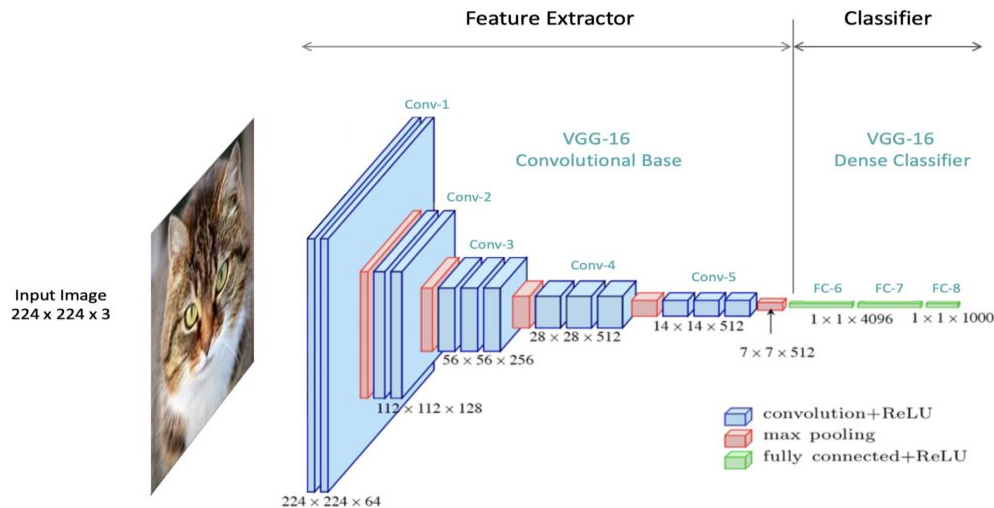


2. Object Detection In Video Analytics

2.1. Introduction

One of the most common applications of deep learning for video analysis is object detection and tracking. This involves detecting and tracking specific objects in a video sequence.

Popular techniques include the use of a convolutional neural network (CNN) to learn complex patterns from data.



2. Object Detection In Video Analytics

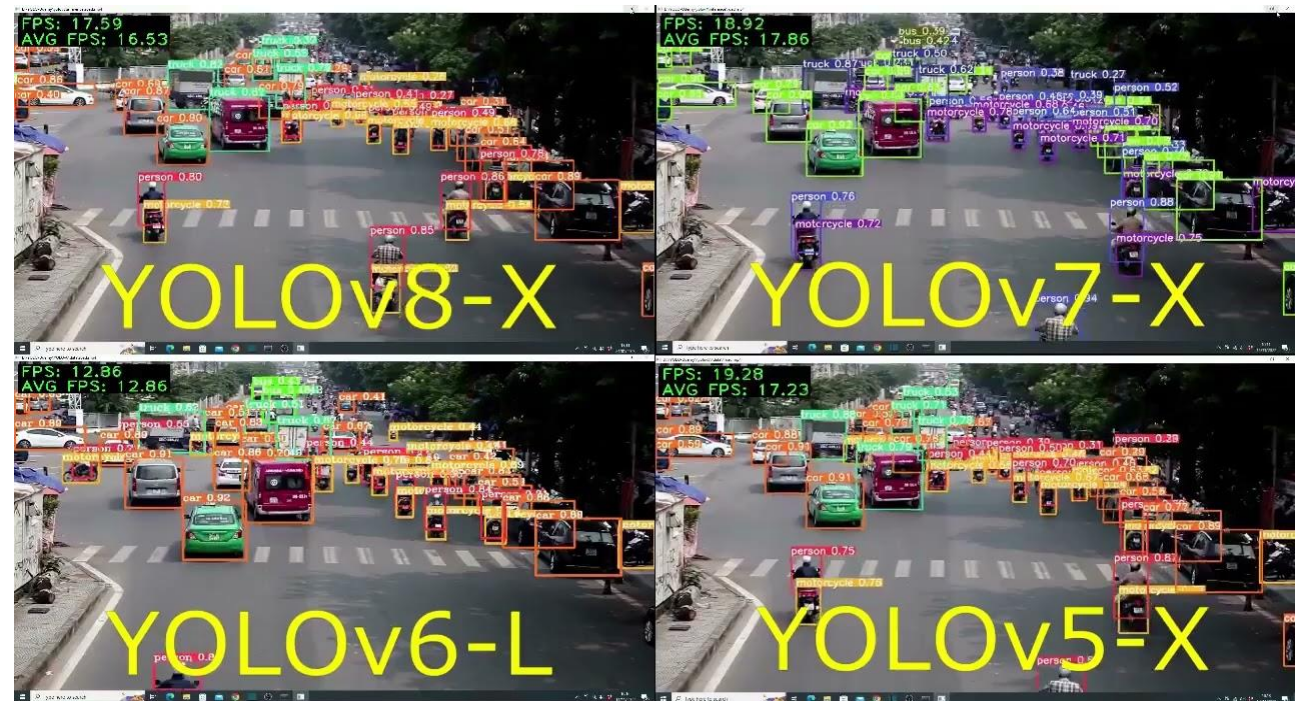
2.2. State of Art Object Detection Model

- Real-time object detection in video streams has been one of the most important
- The most popular algorithms to detect object in video include:

Mask R-CNN, yolov5, yoloR, yolov7, yolov8 (custom trained with collectet video data MS COO

names:

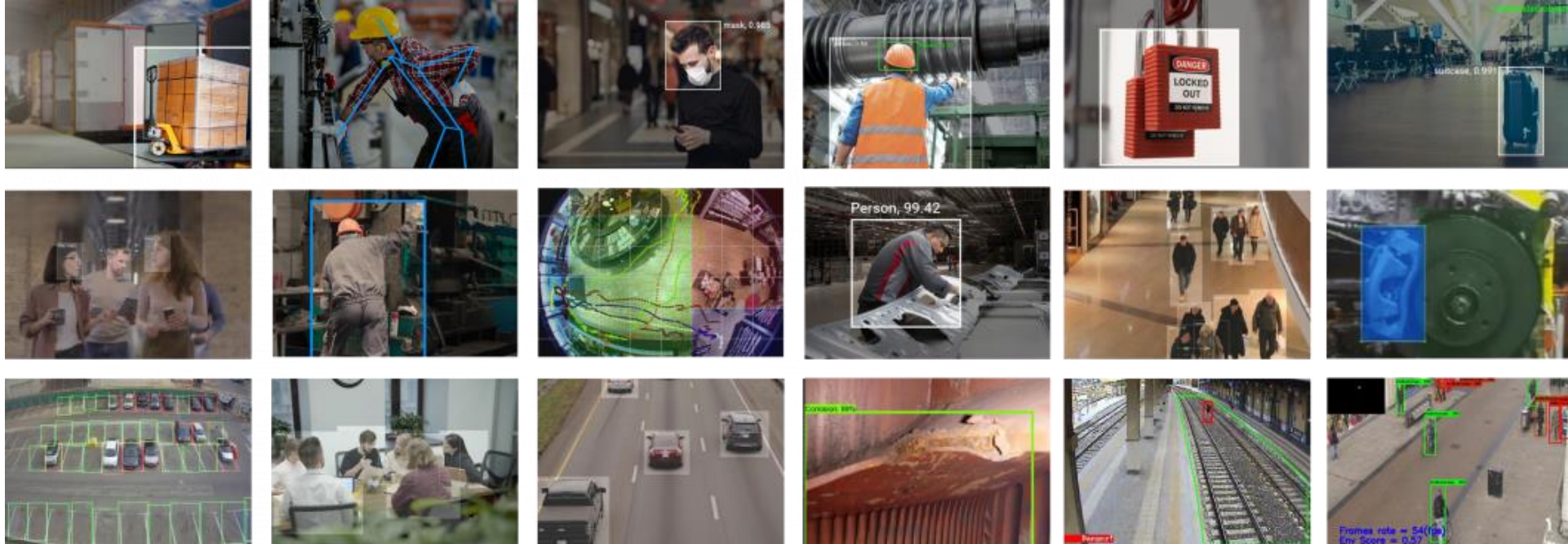
0: Person	352: Dumpling
1: Sneakers	353: Oyster
2: Chair	354: Table Tennis paddle
3: Other Shoe	355: Cosmetics Brush/Eyeline Pencil
4: Hat	356: Chainsaw
5: Car	357: Eraser
6: Lamp	358: Lobster
7: Glasses	359: Durian
8: Bottle	360: Okra
9: Desk	361: Lipstick
10: Cup	362: Cosmetics Mirror
	363: Curling
	364: Table Tennis



3. Video Analytics Application

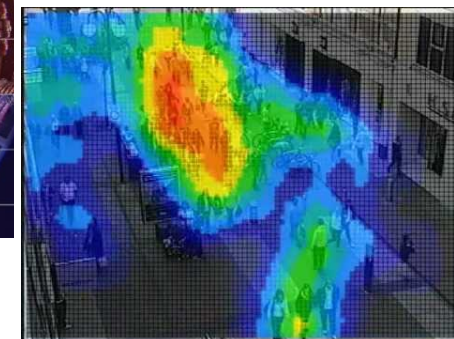
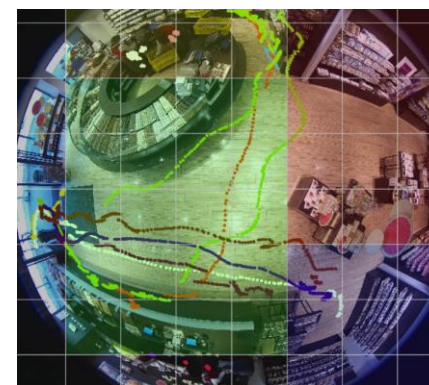
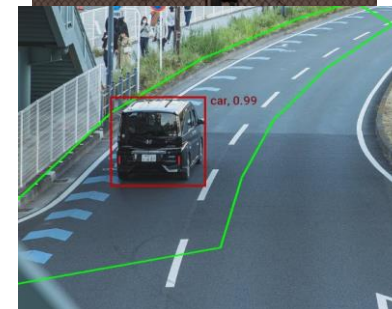
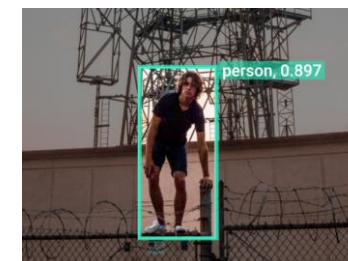
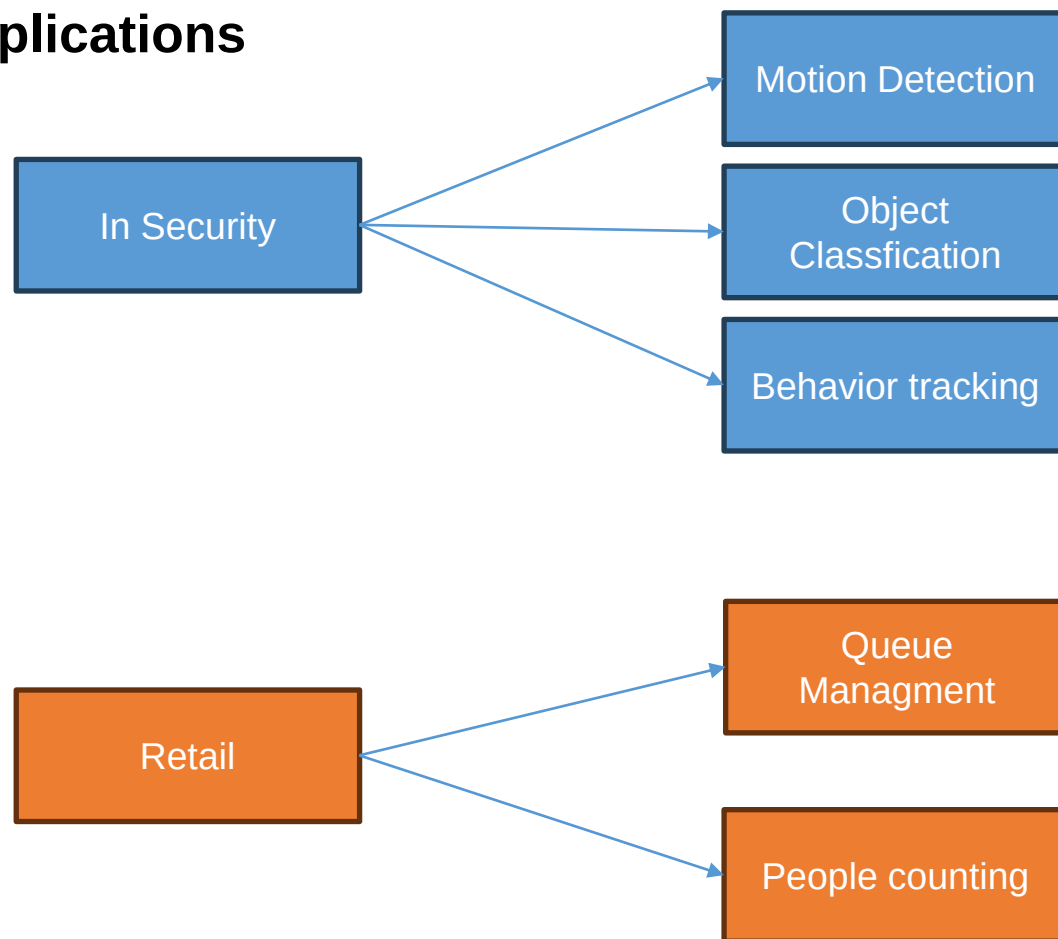
1. Introduction

The largest applications in the video analytics market involve security: incident detection, intrusion management, people counting, traffic monitoring, Automatic Number Plate Recognition (ANPR), facial recognition, AR, ego-motion estimation. In addition, video analytics has been useful for industries such as manufacturing, security, retail, healthcare and hospitality, and more.



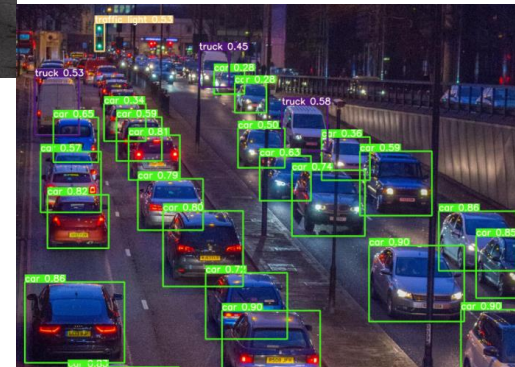
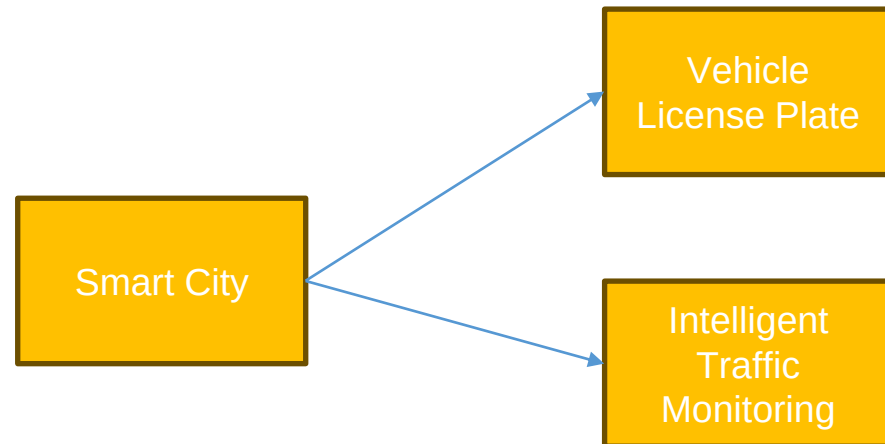
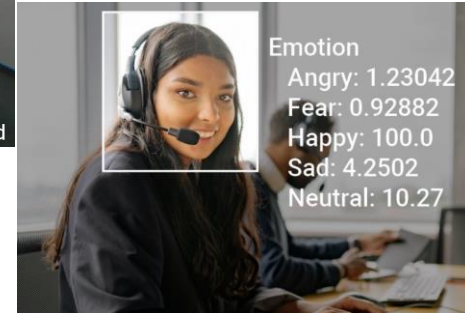
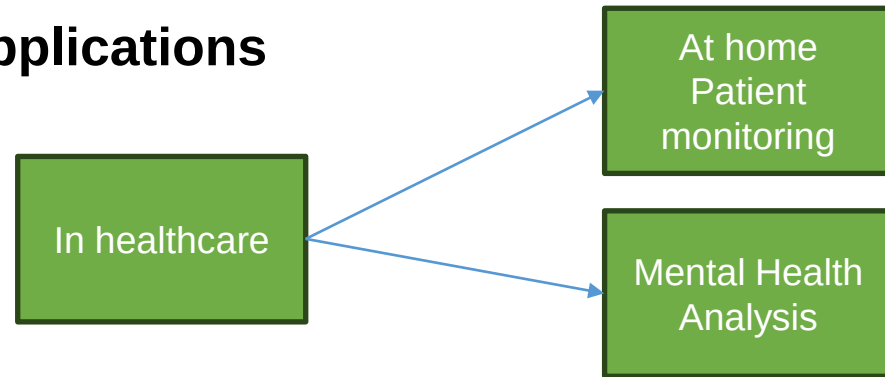
3. Video Analytics Application

2. Applications



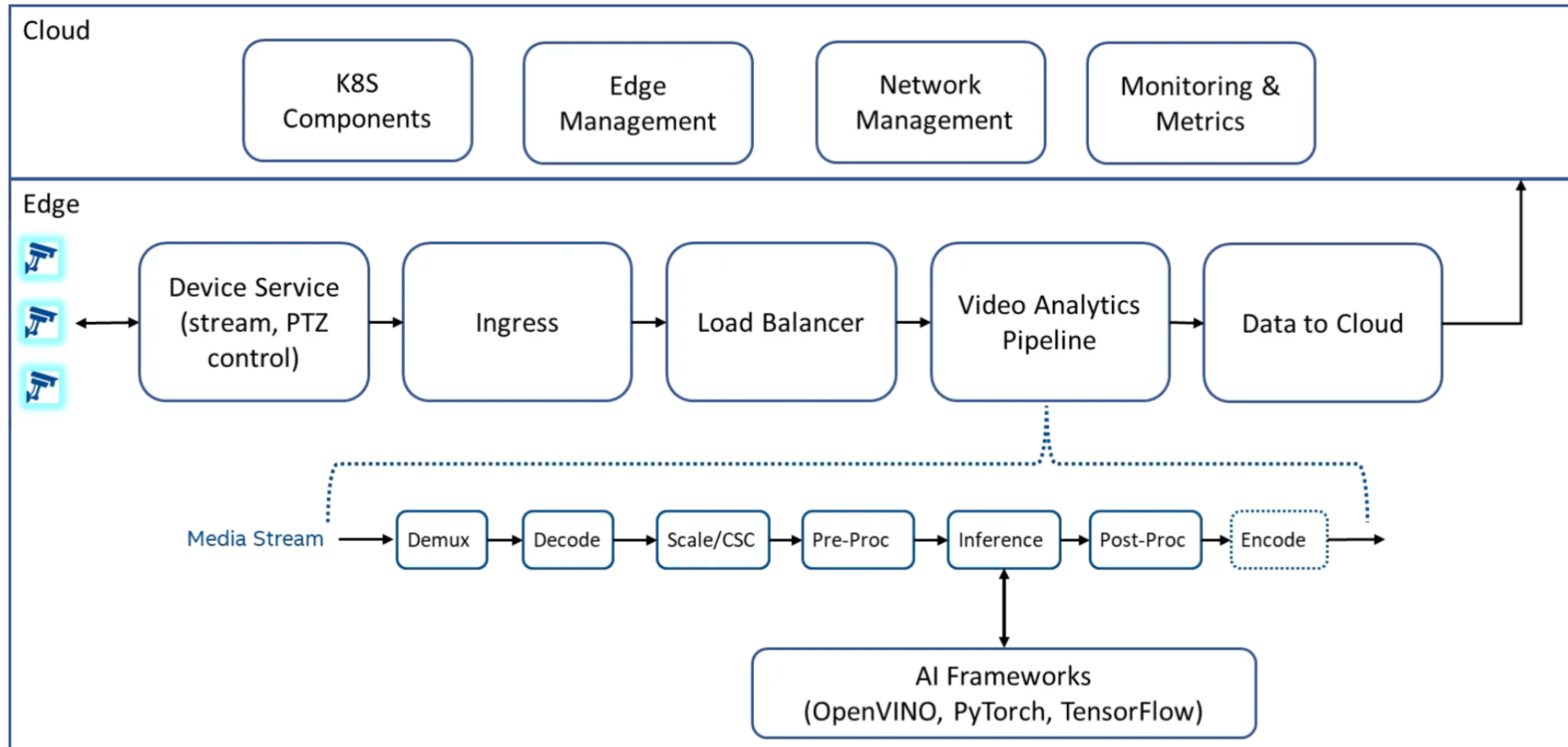
3. Video Analytics Application

2. Applications



4. Demo A Video Analytic Pipeline

1. Video Analytic Pipeline

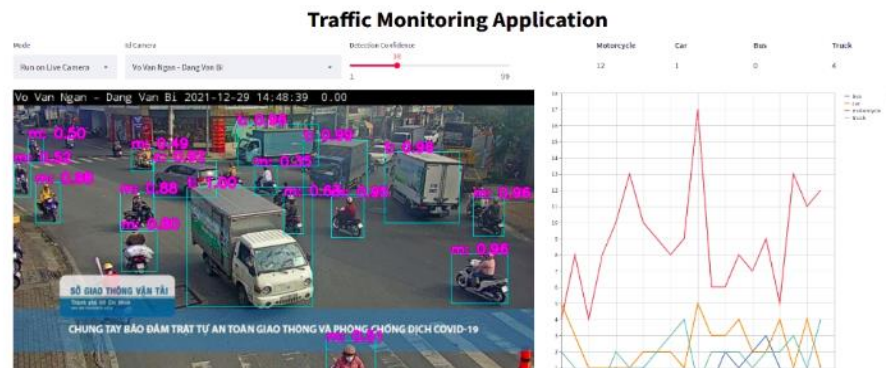


4. Demo A Video Analytic Pipeline

2. Demo HCM Traffic Monitoring

Detect Vehicles

- From Image
- From Video
- From Live Camera



Visualize Information With Charts

Using Realtime Information To Draw Charts

5. Project Lab

1. Requirement

Problem: In a store, they need to collect statistics on the number of customers coming in and out every day and every month and need to know which store attracts more customers based on the number of customers gathering around it.
Let's use video analytics with deep learning to build an application to solve this problem.

Description: Main tasks:

- Counting People cross in line
 - Counting people in zone
 - Heatmap
 - Web UI
- => 1 model people
1 model tracking people

Technique:

Using a state of art object detection model: [Yolov8](#)

Data: [store-video](#)

Deadline: 01/03



THANK YOU!



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